A Literature-Implied Affirmative Answer to the Strict-Slow-Dimension-Growth Question

Math discovery packet

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Status and source question

This is a lightweight *literature-implied answer*, not an original proof. Elliott and Toms close arXiv:0704.1803 with four questions. Question (i), on page 14 of the locally compiled source PDF, asks:

> When do natural examples of simple separable amenable C^*-algebras satisfy one or more of the regularity properties of Section 3? In particular, do simple unital inductive limits of recursive subhomogeneous algebras have strict comparison whenever they have strict slow dimension growth?

Question (i) has an affirmative answer in the later literature.

The decisive later theorem

Andrew S. Toms, *Comparison theory and smooth minimal C^*-dynamics*, arXiv:0805.1688, states as Theorem 1.1:

> Let (A_i, ϕ_i) be a direct sequence of recursive subhomogeneous C^*-algebras with slow dimension growth. Suppose that the limit algebra A is unital and simple. It follows that A has strict comparison of positive elements.

The result is proved again as Theorem 5.3 of that paper.

This applies directly to the source question. For recursive subhomogeneous systems, strict slow dimension growth is the stronger variant of slow dimension growth: in the usual definition, slow dimension growth permits the image of a projection at a fibre to have rank zero or sufficiently large rank, while strict slow dimension growth removes the zero-rank alternative for every nonzero projection. Hence

$$\text{strict slow dimension growth} \quad \Longrightarrow \quad \text{slow dimension growth.}$$

The source question assumes precisely the remaining hypotheses of Theorem 1.1: the algebra is a unital simple inductive limit of recursive subhomogeneous algebras. The theorem therefore yields strict comparison.

Provenance and scope

This is classified as `literature_implied_answer (Question (i), affirmative)`. The later paper cites the Elliott–Toms survey, but it does not explicitly advertise Theorem 1.1 as an answer to the survey's numbered question; the identification is the direct implication above.

The packet resolves only Question (i). It makes no claim about the source paper's Questions (ii)–(iv).

References

- G. A. Elliott and A. S. Toms, *Regularity properties in the classification program for separable amenable C^*-algebras*, arXiv:0704.1803.
- A. S. Toms, *Comparison theory and smooth minimal C^*-dynamics*, arXiv:0805.1688, Theorem 1.1 (also Theorem 5.3).
- N. C. Phillips, *Cancellation and stable rank for direct limits of recursive subhomogeneous algebras*, Trans. Amer. Math. Soc. 359 (2007), 4625–4652, Definition 1.1 (slow and strict slow dimension growth).